

part 2 code finalized-Copy2

December 4, 2025

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[3]: from pylab import *
      from scipy.optimize import curve_fit
      from scipy.integrate import solve_ivp
```

0.1 1MN + 2H2 -> 1MT

0.2 1MN + 2H2 -> 5MT

0.3 5MT -> 1MT

0.4 1MT + 2H2 -> Cis-1MD

0.5 5MT + 2H2 -> Trans-1MD

```
[4]: k = array([1,2.41,0,.11,0,.94,.62,0,0,.01]) #h^-1
      T = 573.15 #K
      v0 = 200 #L
      CMNO = 7 #mol/L
      CH2O = 1.05 #mol/L
      CTO = CMNO + CH2O #mol/L
      p0 = 49.3 #atm
      FTO = 1000 # mol/h
      VO = FTO/CTO #L/h
      FMNO = VO*CMNO #mol/h
      FH2O = CH2O*VO #mol/h
      rhocat = .475 #kg/L
      W0 = rhocat*v0
      print(FMNO)
      print(FH2O)
```

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869.5652173913043
130.43478260869566
```

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[5]: def odefun(W,u):
      FMN, F1MT, F5MT, Ft1MD, Fc1MD = u
      FT = FMN + FH2O + F1MT + F5MT + Ft1MD + Fc1MD

      CMN = FMN/VO #Concentration for liquid, how does change in pressure fit
      ↪into this?
      C1MT = F1MT/VO
```

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C5MT = F5MT/V0
Ct1MD = Ft1MD/V0
Cc1MD = Fc1MD/V0

rMN = -(k[0]+k[1])*CMN
r1MT = k[0]*CMN + k[3]*C5MT - (k[2] + k[3] + k[5]) * C1MT
r5MT = k[1]*CMN + k[3]*C1MT-(k[3]+k[6]+k[7])*C5MT
rc1MD = k[5] * C1MT + k[7]*C5MT + k[8] * Ct1MD - k[9] * Cc1MD
rt1MD = k[4]*C1MT + k[6]*C5MT + k[9]*Ct1MD - k[8]*Cc1MD

dFMNdW = rMN/rhocat
dF1MTdW = r1MT/rhocat
dF5MTdW = r5MT/rhocat
dFc1MDdW = rc1MD/rhocat
dFt1MDdW = rt1MD/rhocat

return [dFMNdW,dF1MTdW,dF5MTdW,dFc1MDdW,dFt1MDdW]

```

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[6]: Wrange = [0, W0]
u0 = array([FMN0, 0, 0, 0, 0])
W = linspace(*Wrange, 100)
result = solve_ivp(odefun, Wrange, u0, t_eval = W)

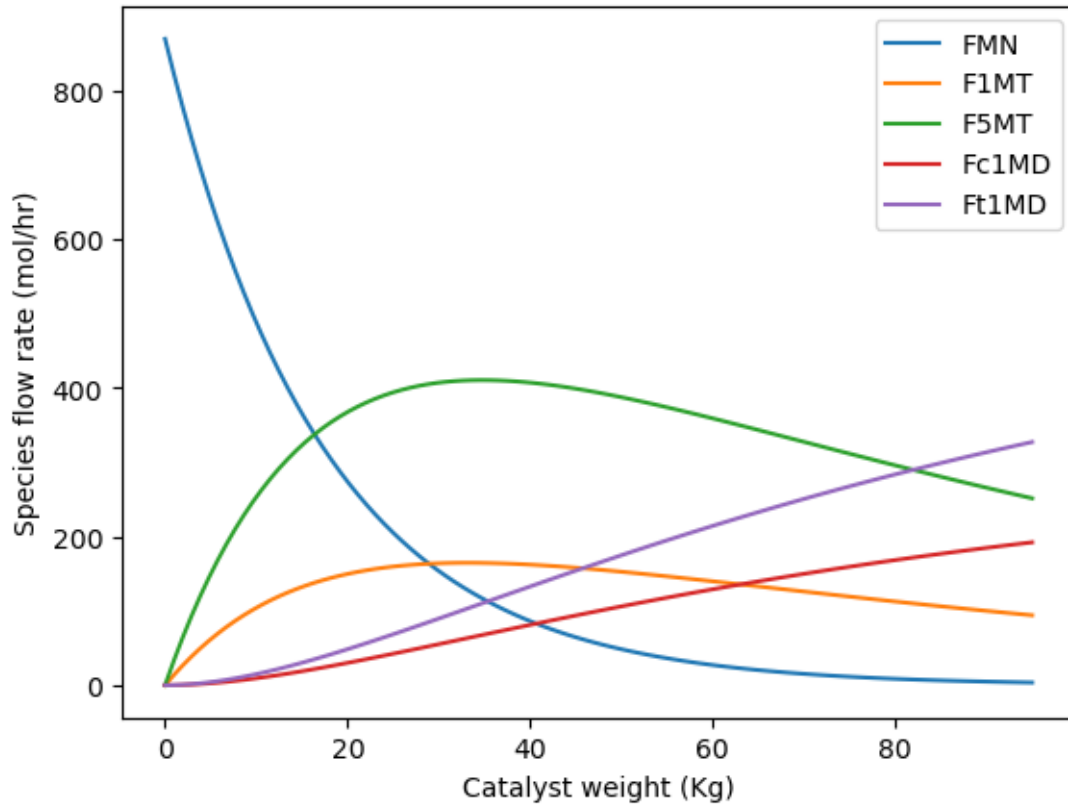
FMN, F1MT, F5MT, Fc1MD, Ft1MD = result.y

```

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[7]: plot(W, FMN, W, F1MT, W, F5MT, W, Fc1MD, W, Ft1MD)
legend(['FMN', 'F1MT', 'F5MT', 'Fc1MD', 'Ft1MD'])
xlabel('Catalyst weight (Kg)')
ylabel('Species flow rate (mol/hr)')
show()

```



```
[8]: idx = argmax(F5MT)
F5MTmax = F5MT[idx]
W5MTmax = W[idx]
F1MTmax = F1MT[idx]
Ft1MDmax = Ft1MD[idx]
Fc1MDmax = Fc1MD[idx]
X = (FMN0-F5MTmax)/FMN0
print(f"W5MTmax = {W5MTmax:.4f} kg")
print(f"F5MTmax = {F5MTmax:.4f} mol/h")
print(f"X = {X:.4f}")
```

```
W5MTmax = 34.5455 kg
F5MTmax = 410.4379 mol/h
X = 0.5280
```

```
[9]: Sf = 1/X #seperation Factor
Wcost = W5MTmax*1000*10.54
CostMN = FMN0 * 142.2 * .24
MH2O = FH2O * 2.016
vH2O = (FH2O*0.0821*T)/p0
CostH2=vH2O*11.4
```

```
Costrectns=(Wcost+CostH2)*8765.82 +CostMN  
print(Costrectns)
```

3204185429.5996623

```
[13]: priceF5MT = F5MTmax*146.23*930.8  
priceF1MT = F1MTmax*146.23*930.8  
pricec1MD = Fc1MDmax*3.09*152.28  
pricet1MD = Ft1MDmax*3.09*152.28  
revenutotal=(priceF5MT+priceF1MT+pricec1MD+pricet1MD)*8765.82  
print(revenutotal)
```

686748975760.4908

```
[11]: M1mn = FMN0 * 142.2 #mass weight  
MH20 = FH20 * 2.016  
  
CostEnergy1mt=FMN0*(573.15-298.15)*224.39*8765.82  
CostEnergyH2=FMN0*(573.15-298.15)*28.84*8765.82  
CostEnergy=(CostEnergy1mt+CostEnergyH2)/1000  
print(CostEnergy)
```

479428612.1373912

```
[18]: Vmax=W5MTmax/rhocat  
print(Vmax)  
CostRctr = 40000 #highballing here because it is extremely small relatively
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72.727272727273

```
[20]: Costperyear = (.1*CostRctr+Costrectns+CostEnergy)*Sf  
profit = revenutotal-Costperyear  
print(profit)
```

679772378991.2449

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[ ]:
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